Northstar Cloud

SaaS Subscription Analytics — Learner Project Brief

1. Engagement Context

You are the analytics workstream supporting the CEO, CFO, CRO and Chief Customer Officer of a multi-product B2B SaaS company. Leadership wants to know whether ARR growth is durable and where management should act first.

Your job is to build a defensible fact base, diagnose value leakage, size opportunities and recommend a 90-day management agenda. The dashboard is evidence; it is not the final answer.

2. Executive Questions

- Where is recurring revenue leaking through churn, contraction or discounting?
- Which customers are at greatest renewal risk, and why?
- Which healthy customers are credible expansion candidates?
- Which acquisition channels create durable customers rather than only cheap leads?
- Is Customer Success capacity aligned to risk-adjusted ARR?
- What 3–5 actions should management prioritize over the next 90 days?

3. New Skills

- MRR/ARR bridge construction: new, expansion, contraction and churn.
- NRR vs GRR and why logo churn alone is incomplete.
- Logo and revenue cohort retention by acquisition vintage.
- Interpretable customer-health scoring.
- Pricing leakage and discount governance.
- CAC combined with retained customer quality.
- Expansion opportunity sizing.
- CSM portfolio capacity and risk-based resource allocation.
- Issue trees, hypotheses, KPI trees, opportunity sizing and answer-first storytelling.

4. Data Estate

Dataset	Scale	Use
Accounts	8,000	Firmographics and customer ownership
Monthly snapshots	145,847	MRR, usage, NPS, movements
Invoices	67,551	Payment behaviour
Support tickets	36,666	Severity, resolution and CSAT
NPS surveys	51,307	Customer advocacy
Marketing spend	270	Channel CAC

5. Required Workflow

Phase	Question	Output

Frame	What decision must management make?	Issue tree and hypotheses
Validate	Can the data be trusted?	DQ log and cleaned layer
Metric layer	Are KPI definitions consistent?	Auditable SaaS KPI layer
Diagnose	What explains the problem?	Cohort/driver fact base
Size	How large is each opportunity?	Risk, expansion and pricing pools
Prioritize	Where should action start?	Impact × confidence × effort
Recommend	What should happen next?	90-day executive memo

6. Technical Deliverables

- PostgreSQL database from raw CSVs.
- 50 SQL questions with commented answers.
- Python EDA with cohorts, health scoring and opportunity sizing.
- Power BI semantic model and 7-page executive report.
- Account-level retention, expansion, pricing and renewal lists.
- Executive memo with low/base/high impact scenarios.

7. Mandatory KPIs

MRR, ARR, Active Logos, ARPA, Logo Retention, NRR, GRR, Expansion MRR, Contraction MRR, Churned MRR, Seat Utilization, NPS, Product Attach Rate, CAC, Pricing Leakage, ARR at Risk and CSM Capacity Utilization.

8. Submission Standard

Each insight should follow: Evidence → So what → Recommendation → Financial impact. Addressable pools must not be presented as guaranteed forecast value.

The company and records are synthetic. The workflow reflects common consulting and SaaS analytics practices and does not reproduce proprietary methods of EY, Bain, McKinsey, or any other firm.